

Technical Limits of AI Explainability: Assessing Faithfulness in Medical Diagnostic Models as a Barrier to Human Replacement

Input Question

Will artificial intelligence replace humans?

Domain

Artificial Intelligence

Validation Status

needs_data

Problem Statement

The question of whether AI will replace humans hinges on the technical capabilities of current AI systems, particularly regarding explainability and handling uncertainty. While AI excels in speed and specific tasks like artifact generation, concerns persist about its ability to provide transparent, faithful explanations for decisions in critical domains like medicine, and its limitation in holistic, intuitive reasoning.

Rationale

Evidence indicates that AI applications in medicine raise significant concerns regarding the explainability of decisions (EV-Q121-0debf3fe850eb4735a3cb609). Furthermore, current computational creativity is largely bounded to artifact generation rather than holistic problem-solving (EV-Q121-718e2bd9e8b12f3b65f90ffc). Theoretical frameworks for AGI exist but lack empirical validation for replacing human intuition in equivocal contexts (EV-Q121-56d39b0cc5e8454b7c6b1218). Therefore, investigating the technical fidelity of AI explainability serves as a proxy for understanding its current limitations in assuming full human responsibility.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Current medical AI systems exhibit a persistent technical deficit in post-hoc explainability fidelity, such that state-of-the-art explanation methods fail to accurately reflect the internal decision logic of high-performance diagnostic models on standard benchmarks.
- **Mechanism**: EV-Q121-0debf3fe850eb4735a3cb609 confirms that concerns have been raised regarding the explainability of AI decisions in medicine. This hypothesis posits that these concerns stem from a fundamental technical misalignment: current post-hoc explanation methods (e.g., SHAP, LIME) approximate model behavior but lack sufficient faithfulness to the actual learned representations in complex deep learning architectures. Consequently, even if model accuracy is high, the 'explanation' remains an unreliable proxy for the true decision

process, validating the technical basis of the cited concerns without asserting unverified regulatory consequences.

- ****Falsifiable Prediction****: If current medical AI possesses sufficient technical explainability to address established concerns, then state-of-the-art post-hoc explanation methods should achieve high faithfulness scores (e.g., >0.9 correlation with ground-truth perturbation effects) and high stability across standard medical imaging benchmarks (e.g., CheXpert). If faithfulness remains significantly below this threshold or degrades in high-confidence predictions, the hypothesis of a persistent technical deficit is supported.
- ****Required Observations****: Quantitative measurement of explanation faithfulness (e.g., insertion/deletion AUC) for SOTA models on CheXpert/MIMIC-CXR ; Comparison of explanation consistency under input perturbation for correct vs. incorrect predictions ; Correlation analysis between model confidence/accuracy and explanation fidelity metrics
- ****Risk of Being Wrong****: Newer explanation methods not covered in current literature might already achieve high fidelity, or the 'concerns' cited in EV-Q121-0debf3fe850eb4735a3cb609 might refer to user-interface issues rather than algorithmic faithfulness, making technical fidelity metrics irrelevant to the actual concern.

Hypothesis 2

- ****Hypothesis****: Current Computational Creativity AI is functionally bounded to artifact generation and lacks demonstrated capability in holistic uncertainty handling, limiting its potential to replace humans in equivocal problem-solving roles.
- ****Mechanism****: EV-Q121-718e2bd9e8b12f3b65f90ffc establishes that current work in Computational Creativity focuses primarily on generating creative artifacts (paintings, poems). This suggests a functional specialization where AI optimizes for output synthesis rather than the adaptive, intuitive reasoning required for equivocality. Combined with the theoretical diversity of AGI approaches (EV-Q121-56d39b0cc5e8454b7c6b1218) which lack empirical validation for intuition, this hypothesis asserts that replacement is currently technically constrained to well-defined generative tasks, leaving holistic/uncertain domains as a verified knowledge gap.
- ****Falsifiable Prediction****: If AI has moved beyond artifact generation to holistic problem solving, then recent Computational Creativity literature must demonstrate validated frameworks for adaptive reasoning in uncertain/equivocal contexts (not just artifact quality). If bibliometric analysis shows >90% of papers still focus exclusively on artifact generation metrics without process-level adaptation benchmarks, the hypothesis of bounded capability is strengthened.
- ****Required Observations****: Bibliometric analysis of Computational Creativity papers (last 5 years) categorizing focus on 'artifact generation' vs. 'adaptive/holistic reasoning' ; Identification of standardized benchmarks for 'intuition' or 'equivocality handling' in current AI creativity research ; Review of AGI framework papers (Logicist/Emergentist) for empirical results on uncertainty handling
- ****Risk of Being Wrong****: Artifact generation might be an emergent substrate for general intelligence (as suggested by Emergentist views in EV-Q121-56d39b0cc5e8454b7c6b1218), meaning the distinction between 'artifact' and 'holistic' processing is artificial and replacement could occur via scaling artifact generators.

Hypothesis 3

- ****Hypothesis****: Theoretical AGI frameworks (Logicist, Emergentist, Universalist) currently lack sufficient empirical grounding to validate claims of human replacement in uncertainty-rich environments.

- **Mechanism**: EV-Q121-56d39b0cc5e8454b7c6b1218 and EV-Q121-906d1928c874b168fdea57f4 outline theoretical approaches to AGI and super-intelligence but exist primarily as abstract concepts in the provided evidence. Without experimental validation linking these theories to measurable performance on uncertainty/equivocality tasks, any claim that they enable human replacement remains speculative. This hypothesis frames the 'replacement' question as currently scientifically undecidable due to the gap between theory and empirical demonstration.
- **Falsifiable Prediction**: If AGI theories are empirically grounded for replacement, then full-text analysis of cited works must contain reproducible experiments demonstrating superior performance on standardized uncertainty benchmarks compared to human baselines. If full-text extraction reveals only theoretical arguments without such empirical validation, the hypothesis of 'premature replacement claims' is supported.
- **Required Observations**: Full-text content extraction and coding of EV-Q121-56d39b0cc5e8454b7c6b1218 and EV-Q121-906d1928c874b168fdea57f4 for empirical results vs. theoretical propositions ; Cross-referencing cited AGI frameworks against established uncertainty-handling benchmarks ; Meta-analysis of citation networks to identify if empirical validations exist in downstream literature
- **Risk of Being Wrong**: Relevant empirical validation may exist in the full text of the papers (which are currently only partially extracted as abstracts/titles) or in newer publications outside the allowed evidence set, making this hypothesis a reflection of extraction limitations rather than true scientific absence.

Technical Details

This experiment validates the technical hypothesis that current post-hoc explainability methods (SHAP, LIME) exhibit low fidelity to the internal decision logic of high-performance medical AI models. The study focuses strictly on the technical metric of 'faithfulness' as a proxy for the 'explainability concerns' cited in EV-Q121-0debf3fe850eb4735a3cb609. We train a ResNet-50 model on the CheXpert dataset for chest X-ray classification. We then apply SHAP and LIME to generate attribution maps. Faithfulness is quantified using the Insertion/Deletion AUC metric, which measures the correlation between the importance assigned by the explanation method and the actual change in model output when features are perturbed. Low faithfulness scores would technically substantiate the basis for the concerns raised in the literature, without making unverified claims about regulatory mandates.

Datasets

Source

```
```json
{
 "name": "CheXpert",
 "description": "A large dataset of chest radiographs with labeled pathologies, used for training and evaluating the diagnostic model.",
 "access_type": "public_restricted",
 "evidence_ids": [
 "EV-Q121-0debf3fe850eb4735a3cb609"
```

]

}

...

Target

```json

{

"name": "Explanation Fidelity Metrics",

"description": "Quantitative metrics (Insertion/Deletion AUC, Stability Score) derived from applying explainability algorithms to the model's predictions. This replaces the invalid 'Synthetic Regulatory Audit Log' with direct technical measurements.",

"access_type": "derived",

"evidence_ids": [

"EV-Q121-0debf3fe850eb4735a3cb609"

]

}

...

Paper Abstract

Background: Artificial intelligence is increasingly deployed in critical domains such as medicine, yet concerns regarding the explainability of its decisions remain prevalent (EV-Q121-0debf3fe850eb4735a3cb609). Whether AI can replace humans depends partly on its ability to provide transparent, trustworthy reasoning. Methods: We investigate the technical fidelity of post-hoc explainability methods (SHAP, LIME) applied to a ResNet-50 model trained on the CheXpert dataset. We measure faithfulness using Insertion/Deletion AUC metrics and stability under perturbation. Validation Plan: The experiment compares explanation fidelity against random baselines and gradient-based saliency maps. Results: pending. This study aims to quantify the technical gap in AI explainability, providing empirical evidence for the limitations of current AI in assuming full human-like decision-making responsibility.

Methods

1. Model Training: Train a ResNet-50 convolutional neural network on the CheXpert training set for multi-label classification of chest pathologies. 2. Explanation Generation: Apply SHAP (DeepExplainer) and LIME to a held-out test set to generate pixel-level attribution maps. 3. Faithfulness Evaluation: Compute Insertion and Deletion AUC curves. For Insertion, pixels are added in order of importance; for Deletion, they are removed. The area under these curves indicates how well the explanation captures the model's reliance on specific features. 4. Stability Analysis: Measure the Jaccard similarity of explanations for original images vs. slightly perturbed versions (Gaussian noise) to assess robustness. 5. Statistical Analysis: Compare mean faithfulness scores against a random baseline to determine if current methods provide statistically significant insight into model logic.

Experiments

Baselines

```
```json
[
 "Random Baseline: Attribution maps generated with random weights to establish the lower bound of faithfulness.",
 "Gradient-based Saliency: Vanilla backpropagation gradients as a standard, non-perturbation-based explanation method."
]
```
```

Metrics

```
```json
[
 "Insertion AUC: Area under the curve for the insertion of top-k important pixels.",
 "Deletion AUC: Area under the curve for the deletion of top-k important pixels.",
 "Explanation Stability: Jaccard index of top-10% important pixels between original and perturbed inputs."
]
```
```

Ablation

Compare performance of SHAP vs. LIME across different pathology classes (e.g., Pneumonia vs. Cardiomegaly) to identify if fidelity varies by disease complexity.

Validation Protocol

5-fold cross-validation on the CheXpert test split. Significance of differences in AUC scores between baselines and proposed methods tested using paired t-tests ($p < 0.05$).

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q121-0debf3fe850eb4735a3cb609** · arxiv · arXiv:2304.04780

- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2304.04780.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=8bf98175fb50470152979c0c07dab71f7c7e47db7cbeb7f2a08ff45fd8e73df0
- ****EV-Q121-718e2bd9e8b12f3b65f90ffc**** · arxiv · arXiv:2204.10358
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2204.10358.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=4d461808343c0fb93cd3b2dda609a279b77a2df3b129027fd05366e51161c17c
- ****EV-Q121-56d39b0cc5e8454b7c6b1218**** · arxiv · arXiv:2110.01831
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2110.01831.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=2a38c1fb43901330898f27b46393ec9cefd2e95a80f0d6cf708430c4e7227cad

Reviewer Comments

- The revised hypothesis correctly pivots from an unverified socio-legal claim to a strictly technical deficit in explainability fidelity, fully addressing the 'Causal Gap' and 'Overclaiming Mechanism' critical issues from the previous review.
- The experiment design now uses valid, reproducible technical metrics (Insertion/Deletion AUC, Stability) on a standard dataset (CheXpert), resolving the 'Invalid Operationalization' and 'Dataset Mismatch' issues. The removal of the 'Synthetic Regulatory Audit Log' eliminates the fictitious target variable.
- Evidence grounding is now robust: the hypothesis is directly supported by EV-Q121-0debf3fe850eb4735a3cb609 regarding 'concerns about explainability', without extrapolating to regulatory mandates.
- Results are correctly marked as pending, and the reproducibility checklist is comprehensive. The plan is now scientifically sound within the bounds of the allowed evidence.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- PyTorch code for ResNet-50 training with fixed random seeds.

- Implementation of SHAP DeepExplainer and LIME for image data.
- Scripts for computing Insertion/Deletion AUC metrics.
- Pre-processing pipeline for CheXpert images (resizing, normalization).
- Detailed log of hyperparameters for model training and explanation generation.